lsm_tree_storage_engines (Initial Draft)

Domain: Database Systems | Document ID: lsm_tree_storage_engines_v1
Specification Standard: IEEE Capstone Evaluation Corpus

Executive Overview:

Log-Structured Merge-tree architecture: in-memory MemTable, Write-Ahead Log (WAL), and multi-level SSTable compaction.

Methodological Verification & System Lineage:

This specification is catalogued in IntentCloud for intent-aware hybrid retrieval.
Text chunks are indexed into 384-dimensional dense semantic vectors and 1,000,003-dimensional sparse feature-hashed token projections.
Candidate pools are unified using scale-invariant Reciprocal Rank Fusion (k=60).
Explainable citation attribution is extracted using cross-encoder cross-attention.